

Lab Nr. 1, Numerical Calculus

Review of Matlab

1. Matlab files .m

- *script*, *live script* and *function* files

2. Vectors and matrices

- generation
- operations (dot and matrix operations)
- special matrices

3. Polynomials

- evaluation, *polyval*
- rootfinding, *roots*

4. Graphics

- *plot*, *plot3* with options
- *subplot*
- *mesh*

5. Display

- *disp*
- *fprintf* with options

Applications

1. Let $p(x) = x^5 - 5x^4 - 16x^3 + 16x^2 - 17x + 21$.

- a) Plot the graph of p on the interval $[-4, 7.2]$.
- b) Compute $p(-2.5)$.
- c) Find the roots of p .

2. Plot the functions $f, g, h : [0, 2\pi] \rightarrow \mathbb{R}$, $f(x) = \sin x$, $g(x) = \sin 2x$, $h(x) = \sin 3x$, on the same figure, in three tiled positions, one on top of the other.

3. For $R, r \in \mathbb{R}_+$, consider the *epicycloid* (also called *hypercycloid*), given by the parametric equations

$$\begin{cases} x(t) &= (R + r) \cos t - r \cos \left(\left(\frac{R}{r} + 1 \right) t \right) \\ y(t) &= (R + r) \sin t - r \sin \left(\left(\frac{R}{r} + 1 \right) t \right) \end{cases}, \quad t \in [0, 10\pi].$$

Plot its graph, for $R = 3.8$, $r = 1$ (see details at <https://en.wikipedia.org/wiki/Epicycloid>).

4. Plot the function of two variables $f : [-2, 2] \times [0.5, 4.5] \rightarrow \mathbb{R}$, $f(x, y) = \sin(e^x) \cos(\ln y)$ (using both *mesh* and *plot3*).

Optional

5. Consider the expression

$$f(n) = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\dots}}}},$$

where $n \in \mathbb{N}$ is the number of fractions (e.g., $f(0) = 1 + 1$, $f(1) = 1 + \frac{1}{1+1} = \frac{3}{2}$). Compute $f(2)$, $f(10)$, $f(100)$ and $f(2025)$. What can be noticed?

Lab Nr. 2, Numerical Calculus

Taylor Polynomials and Taylor Series

• Short theory review

Theorem 1. [Taylor's Theorem] Let $f : [a, b] \rightarrow \mathbb{R}$ be a function with $n + 1$ continuous derivatives on $[a, b]$, for some $n \geq 0$ and let $x, x_0 \in [a, b]$. Then

$$\begin{aligned} f(x) &= T_n f(x) + R_{n+1}(x), \text{ with} \\ T_n f(x) &= f(x_0) + \frac{(x - x_0)}{1!} f'(x_0) + \dots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) \text{ and} \\ R_{n+1}(x) &= \frac{1}{n!} \int_{x_0}^x (x - t)^n f^{(n+1)}(t) dt = \frac{(x - x_0)^{n+1}}{(n + 1)!} f^{(n+1)}(\xi), \xi \text{ between } x \text{ and } x_0. \end{aligned} \quad (1)$$

Using Taylor's theorem with $x_0 = 0$, we obtain MacLaurin's formula:

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \frac{x^{n+1}}{(n + 1)!} f^{(n+1)}(\xi), \xi \text{ between } x \text{ and } x_0.$$

MacLaurin's formula for some common functions:

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n + 1)!} e^{\xi_x}, \quad (2)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + (-1)^{n+1} \frac{x^{2n+2}}{(2n + 2)!} \cos \xi_x, \quad (3)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n - 1)!} + (-1)^n \frac{x^{2n+1}}{(2n + 1)!} \sin \xi_x, \quad (4)$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^{n-1} \frac{x^n}{n} + (-1)^n \frac{x^{n+1}}{n + 1} \frac{1}{(1 + \xi_x)^{n+1}}, \quad (5)$$

$$(1 + x)^a = 1 + \binom{a}{1} x + \binom{a}{2} x^2 + \dots + \binom{a}{n} x^n + \binom{a}{n + 1} \frac{x^{n+1}}{(1 + \xi_x)^{n+1-a}}, \quad (6)$$

$$\binom{a}{k} = \frac{a(a - 1) \cdots (a - k + 1)}{k!}, \quad k = 1, 2, 3, \dots, a \in \mathbb{R}.$$

As $n \rightarrow \infty$, the Taylor series on the right-hand-sides of (2) – (4) converge for $x \in \mathbb{R}$, and those in (5) – (6), for $|x| < 1$.

• **Matlab function** *taylor*

Applications

1. Use formula (2) to do the following:

- a) Graph on the same set of axes the function e^x and its Taylor polynomials of degrees 1, 2, 3 and 4, for $x \in [-3, 3]$;
- b) Approximate e with 6 correct decimals.

2. Using formula (4), do the following:
- a) Graph on the same set of axes the function $\sin x$ and its Taylor polynomials of degrees 3 and 5, for $x \in [-\pi, \pi]$;
 - b) Approximate $\sin\left(\frac{\pi}{5}\right)$ with 5 correct decimals;
 - c) What happens for $x = \frac{10\pi}{3}$? Explain the phenomenon and find a solution.
- 3.
- a) Using formula (5), graph on the same set of axes the function $\ln(1+x)$ and its Taylor polynomials of degrees 2 and 5, for $x \in [-0.9, 1]$;
 - b) How many terms would be necessary in (5) to approximate $\ln 2$ with 5 correct decimals?
 - c) Use formula (5) to find a Taylor series expansion for the function $\ln(1-x)$.
 - d) Use formula (5) and the one found in part c) to derive a Taylor series expansion for the function $\ln \frac{1+x}{1-x}$.
 - e) Use the formula found in part d) to approximate $\ln 2$ with 5 correct decimals. How many terms are necessary?

Optional

4. Use formula (6) to approximate $\sqrt[3]{999}$ with 10 correct decimals (**Caution!!** The series in (6) converges *only* for $|x| < 1$).

Lab Nr. 3, Numerical Calculus

Divided and Finite Differences

1. Write a MATLAB function that generates the divided differences table, given the nodes and the function values (i.e., for *simple* nodes).
2. Write a MATLAB function that generates the divided differences table, given the nodes, the function and the derivative values (i.e., for *double* nodes).
3. Write MATLAB functions that generate *forward* and *backward* finite differences tables, given the function values (at equidistant nodes).

Applications

1. Let $f : (-1, \infty) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{1}{1+x}$.

- a) Find the divided differences table of f at the simple nodes $x_0 = 0, x_1 = 1, x_2 = 2$;
- b) Find the divided differences table of f at the double nodes $x_0 = 0, x_1 = 1, x_2 = 2$;
- c) Repeat parts **a)** and **b)** at 11 equidistant nodes on the interval $[1, 2]$.

2. The following data is gathered:

x	-2	-1	0	1	2	3	4
f	-5	1	1	1	7	25	60

- a) Construct the divided differences table of the data;
- b) Find the forward differences table of the data;
- c) Find the backward differences table of the data.

Lab Nr. 4, Numerical Calculus

Linear Systems I

Direct Methods: Gaussian Elimination, Factorizations

1. Implement back substitution for solving an upper triangular linear system and forward substitution for solving a lower triangular linear system.
2. Implement Gaussian elimination with partial pivoting.
3. Implement a routine that solves the system $Ax = b$ using
 - LUP factorization;
 - Cholesky factorization;
 - QR factorization.

Applications

1. Use both methods (Gaussian elimination and factorizations, when possible) to solve the linear system

$$\begin{bmatrix} 2 & 1 & -1 & -2 \\ 4 & 4 & 1 & 3 \\ -6 & -1 & 10 & 10 \\ -2 & 1 & 8 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ -5 \\ 1 \end{bmatrix}.$$

2. Use both methods (Gaussian elimination and factorizations, when possible) to solve the following general linear system, that can be generated of any order $n \geq 3$:

$$\begin{cases} 5x_1 - x_2 = 4 \\ -x_{j-1} + 5x_j - x_{j+1} = 3, \quad j = \overline{2, n-1} \\ -x_{n-1} + 5x_n = 4 \end{cases}$$

Optional

3. Use both methods (Gaussian elimination and factorizations, when possible) to solve the following general linear system, that can be generated of any order $n \geq 7$:

$$\begin{bmatrix} 5 & -1 & 0 & -1 & \dots & \dots & 0 \\ -1 & 5 & -1 & 0 & -1 & \dots & \dots & \vdots \\ 0 & -1 & 5 & -1 & \ddots & \ddots & \vdots & \vdots \\ -1 & & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 & 5 & -1 & 0 & -1 \\ 0 & \dots & \ddots & 0 & -1 & 5 & -1 & 0 \\ 0 & \dots & & -1 & 0 & -1 & 5 & -1 \\ 0 & \dots & & & -1 & 0 & -1 & 5 \end{bmatrix} x = \begin{bmatrix} 3 \\ 2 \\ 2 \\ 1 \\ \vdots \\ 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}.$$

Lab Nr. 5, Numerical Calculus

Linear Systems II

Iterative Methods: Jacobi, Gauss-Seidel, SOR; Conditioning of Linear Systems

Implement the Jacobi and Gauss-Seidel iterative methods, in the form

$$[x, nit] = \text{Name}(A, b, x0, err, maxnit),$$

where

A is the system matrix,

b is the right-hand side of the system,

$x0$ is the initial solution,

err is the desired error,

$maxnit$ is the maximum number of iterations allowed (in case the method does not converge),

x is the iterative solution of the system,

nit is the number of iterations.

Applications

1. Use both iterative methods to approximate the solution of the linear system (Problem 1, Lab Nr. 4)

$$\begin{cases} 5x_1 - x_2 & = 4 \\ -x_{j-1} + 5x_j - x_{j+1} & = 3, \quad j = \overline{2, n-1}, \\ -x_{n-1} + 5x_n & = 4 \end{cases}$$

for $n = 1000$. Compare the number of iterations needed for an error of 10^{-5} .

2. (Wilson's Example) Consider the system $Ax = b$, where

$$A = \begin{bmatrix} 10 & 7 & 8 & 7 \\ 7 & 5 & 6 & 5 \\ 8 & 6 & 10 & 9 \\ 7 & 5 & 9 & 10 \end{bmatrix}, \quad b = \begin{bmatrix} 32 \\ 23 \\ 33 \\ 31 \end{bmatrix}.$$

- a) Solve the system.

- b) Perturb b to $\tilde{b} = \begin{bmatrix} 32.1 \\ 22.9 \\ 33.1 \\ 30.9 \end{bmatrix}$. Solve the system $Ax = \tilde{b}$. What are the input and output relative errors?

- c) The same question for the perturbed system matrix $\tilde{A} = \begin{bmatrix} 10 & 7 & 8.1 & 7.2 \\ 7.8 & 5.04 & 6 & 5 \\ 8 & 5.98 & 9.89 & 9 \\ 6.99 & 4.99 & 9 & 9.98 \end{bmatrix}$.

- d) Explain the phenomenon.

Optional

3. Implement the successive overrelaxation (SOR) method, in the form

$$[x, nit] = Name(A, b, \omega, x0, err, maxnit),$$

where ω is the optimal relaxation parameter. Then use it to approximate the solution of the system (Problem 3, Lab Nr. 4)

$$\begin{bmatrix} 5 & -1 & 0 & -1 & \dots & \dots & 0 \\ -1 & 5 & -1 & 0 & -1 & \dots & \dots & \vdots \\ 0 & -1 & 5 & -1 & \ddots & \ddots & \vdots & \vdots \\ -1 & & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 & 5 & -1 & 0 & -1 \\ 0 & \dots & \ddots & 0 & -1 & 5 & -1 & 0 \\ 0 & \dots & & -1 & 0 & -1 & 5 & -1 \\ 0 & \dots & & & -1 & 0 & -1 & 5 \end{bmatrix} x = \begin{bmatrix} 3 \\ 2 \\ 2 \\ 1 \\ \vdots \\ 1 \\ 2 \\ 2 \\ 3 \end{bmatrix},$$

for $n = 10$ and $n = 1000$. Compare the number of iterations needed in all three iterative methods, for a precision of 10^{-5} .

Lab Nr. 6, Numerical Calculus
Lagrange Interpolation I
Fundamental Polynomials; Barycentric Formula

1. Implement Lagrange interpolation, using the classical form (with Lagrange basis polynomials).
2. Implement Lagrange interpolation, using the barycentric formula.

Applications

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \frac{x+1}{3x^2+2x+1}.$$

- a) Find the Lagrange polynomial $L_9 f$ (in classical form) that interpolates f at 10 equally spaced nodes in $[-2, 4]$;
 - b) Plot the nodes, f and $L_9 f$, on the same set of axes. What can be noticed?
 - c) Plot the errors $|f - L_9 f|$ and compute the maximum error on $[-2, 4]$.
 - d) Approximate $f(1/2)$ by $(L_9 f)(1/2)$. What is the error of this approximation?
2. According to the International Data Base of the U.S. Census Bureau, the following data represents the population of the world between 1980 and 2020:

Year	1980	1990	2000	2010	2020
Pop. (mln.people)	4451	5287	6090	6970	7821

Use the barycentric formula to interpolate these data and approximate the world population in 2005 and 2015. Knowing the actual world population was 6474 million in 2005 and 7405 million people in 2015, how good are these approximations (compute the relative errors)?

3. Use Lagrange interpolation (in either form) to approximate $\sqrt{118}$ with 3 correct decimals.

Optional

4. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$f(x) = |x| + \frac{1}{2}x - x^2.$$

- a) Plot on the same set of axes the function f and its Lagrange interpolant at $m = 10$ equidistant nodes in $[-1, 1]$. Then at $m = 20$ nodes. What do you notice?
- b) Repeat point a) for $m = 20$ and $m = 100$ Chebyshev nodes of the first kind.
- c) Repeat point b) for $m = 20$ and $m = 100$ Chebyshev nodes of the second kind.

Lab Nr. 7, Numerical Calculus

Lagrange Interpolation II

Newton Form; Aitken's Algorithm

1. Implement Lagrange interpolation, in Newton's divided differences form.
2. Implement Lagrange interpolation, using Aitken's algorithm.

Applications

1. (This is a good problem to also do "by hand".) Consider the function $f(x) = \cos(\pi x)$ and the nodes $\{0, 1/3, 1/2, 1\}$.
 - a) Find the Lagrange polynomial $L_3 f$ using Newton's divided differences;
 - b) Find a bound for the error $R_3 f$;
 - c) Plot f and $L_3 f$, on the same set of axes, for $x \in [0, 1]$.
 - d) Use $L_3 f$ to approximate $\cos\left(\frac{\pi}{5}\right)$;
 - e) Find a bound for the error of this approximation.
2. Approximate $\sqrt{2}$ using Aitken's algorithm to interpolate the function $f(x) = 2^x$ at 9 equidistant nodes on the interval $[-4, 4]$.
3. The following table contains values of $\lg x$ ($= \log_{10} x$) rounded to 7 decimals:

x	$\lg x$
1000	3.0000000
1010	3.0043214
1020	3.0086002
1030	3.0128372
1040	3.0170333
1050	3.0211893

Use Lagrange interpolation with divided differences to approximate $\lg x$, for $x = 1001, 1002, \dots, 1009$.

Optional

4. The following table contains values of $\sin x$ at equally spaced nodes (given in degrees, not radians):

x	$\lg x$
39°	0.6293204
41°	0.6560590
43°	0.6819984
45°	0.7071068
47°	0.7313597
49°	0.7547096
51°	0.7771460

Use Newton interpolation with *forward differences* to approximate $\sin 40^\circ$, $\sin 44^\circ$ and $\sin 50^\circ$.

Lab Nr. 8, Numerical Calculus

Hermite Interpolation

Summary and Review of Polynomial Interpolation

Implement Hermite interpolation with double nodes, using divided differences.

Applications

1. Consider the function $f : (-1, \infty) \rightarrow \mathbb{R}$, $f(x) = \frac{1}{1+x}$ and the nodes $x_0 = 0$, $x_1 = 1$ and $x_2 = 2$. Plot on the same set of axes the Lagrange and the Hermite (with double nodes) polynomials that interpolate the data. What are the degrees of the two polynomials?
2. The data below refers to a moving object.

Time	0	3	5	8	13
Distance	0	225	383	623	993
Velocity	0	77	80	74	72

Use interpolation to approximate the position and the speed of the object at time $t = 10$.

Optional

3. Approximate $\sqrt[3]{7}$ using Hermite interpolation with 4 double nodes.

Review of Polynomial Interpolation

I. Summary

1. What **type** of interpolation? It depends on what information is given:

- **only** values of the function are given, e.g. $f(-1), f(0), f(1) \rightarrow$ **Lagrange**
- values of the function are known for **all** nodes and, in addition, for some nodes, **all the derivatives up to some order (not the same for all nodes)** are also given, e.g. $f(-1), f(0), f'(0), f''(0), f(1), f'(1) \rightarrow$ **Hermite**
- **random** values of the **function and/or of the derivatives** are given, e.g. $f(-1), f'(0), f''(0), f'(1), f'''(2) \rightarrow$ **Birkhoff**

2. What **computational formula** is more convenient?

- Lagrange $\rightarrow$ **Newton's divided (/forward/backward) differences** or Aitken's algorithm
- Hermite $\rightarrow$ **Newton's divided differences** (**caution!!**, for multiple nodes, divided differences are computed with *derivatives*)
- Birkhoff $\rightarrow$ **fundamental (basis) polynomials or directly** (determine the coefficients of the polynomial from the interpolation conditions)

3. The **degree** of the interpolation polynomial

- the **set of indices** I_k consists of the **orders** of the derivatives that are given for each node x_k (the derivative of order 0 is the function itself), e.g.

$$\begin{aligned}x_0 &= -1 \text{ and we know } f(-1), f'(-1) \rightarrow I_0 = \{0, 1\} \\x_1 &= 0 \text{ and we know } f'(0), f''(0) \rightarrow I_1 = \{1, 2\} \\x_2 &= 1 \text{ and we know } f(1), f'''(1), f^{(iv)}(1) \rightarrow I_2 = \{0, 3, 4\} \\x_3 &= 2 \text{ and we know } f''(2), f^{(v)}(2) \rightarrow I_3 = \{2, 5\}\end{aligned}$$

- the **degree** of the interpolation polynomial with nodes $x_0, \dots, x_m$ is

$$n = |I_0| + \dots + |I_m| - 1,$$

where $|A| = \text{card}(A)$ is the number of elements of the set A .

4. Birkhoff interpolation

- The **notation** for **Birkhoff fundamental polynomials**: b_{ij} , where i is the index of the **node**, j is the **order of the derivative** that is known for that node, e.g. if we are given

$$f'(-1), f'(0), f''(0), f(1), f''(1),$$

that means

- $f'(-1)$: for node $x_0 = -1$, the **first** derivative is known $\rightarrow b_{01}$
- $f'(0), f''(0)$: for node $x_1 = 0$, the **first** and **second** derivatives are given $\rightarrow b_{11}, b_{12}$

- $f(1)$, $f''(1)$: for node $x_2 = 1$, the **zero** and **second** derivatives are given $\rightarrow b_{20}$, b_{22} .
So the fundamental polynomials are

$$b_{01}, b_{11}, b_{12}, b_{20}, b_{22}$$

and the pairs of indices used are

$$(0, 1), (1, 1), (1, 2), (2, 0), (2, 2). (*)$$

- The **interpolation polynomial** is then the **linear combination** of the basis polynomials found above, each having as coefficient the value of the function/derivative that it corresponds to:

$$(Bf)(x) = b_{01}f'(-1) + b_{11}f'(0) + b_{12}f''(0) + b_{20}f(1) + b_{22}f''(1).$$

- Each basis polynomial b_{ij} has (at most) **the same degree** as the overall interpolation polynomial Bf (discussed above) and its coefficients are determined from the conditions

$$b_{ij}^{(j)}(x_i) = 1,$$

while at **all other combinations of indices used in the writing of the polynomial, it is 0**. For the example above (check (*)):

$$\left\{ \begin{array}{l} (0, 1) \rightarrow b_{01}'(x_0) = 1 \\ (1, 1) \rightarrow b_{01}'(x_1) = 0 \\ (1, 2) \rightarrow b_{01}''(x_1) = 0 \\ (2, 0) \rightarrow b_{01}(x_2) = 0 \\ (2, 2) \rightarrow b_{01}''(x_2) = 0 \end{array} \right\}, \left\{ \begin{array}{l} (0, 1) \rightarrow b_{11}'(x_0) = 0 \\ (1, 1) \rightarrow b_{11}'(x_1) = 1 \\ (1, 2) \rightarrow b_{11}''(x_1) = 0 \\ (2, 0) \rightarrow b_{11}(x_2) = 0 \\ (2, 2) \rightarrow b_{11}''(x_2) = 0 \end{array} \right\}, \left\{ \begin{array}{l} (0, 1) \rightarrow b_{12}'(x_0) = 0 \\ (1, 1) \rightarrow b_{12}'(x_1) = 0 \\ (1, 2) \rightarrow b_{12}''(x_1) = 1 \\ (2, 0) \rightarrow b_{12}(x_2) = 0 \\ (2, 2) \rightarrow b_{12}''(x_2) = 0 \end{array} \right\},$$

and so on.

- If finding the Birkhoff polynomial **directly**, just impose the interpolation conditions on Bf (after determining its degree **correctly**!). There is just one system in this case, but it is, in general, **more difficult** to solve than the systems for b_{ij} and the remainder is more difficult to find.
5. **Check** at the end that the polynomial found **satisfies** the interpolation conditions. If it exists, the interpolating polynomial is **unique**.

II. Peano's Theorem for the Remainder

- We have an approximating formula

$$\begin{aligned} f(x) &\approx (P_n f)(x), \text{ or} \\ f(x) &= (P_n f)(x) + (R_n(f))(x), \text{ for } x \in [a, b], \end{aligned}$$

where $[a, b]$ is the smallest interval containing all the interpolation nodes. The formula has **degree of precision (or exactness)** $d = n$, if it is **exact** for **all** polynomials of degree up to n , i.e.

$$\begin{aligned} f(x) &= (P_n f)(x), \text{ for } f(x) = e_k(x) = x^k, \quad k = 0, 1, \dots, n, \\ e_{n+1}(x) &\neq (P_n e_{n+1})(x), \text{ or, equivalently, } R_n(e_{n+1}) \neq 0. \end{aligned}$$

- The remainder has the form

$$(R_n(f))(x) = \int_a^b K_n(x, t) f^{(n+1)}(t) dt$$

where

$$K_n(x, t) = R_n\left(\frac{(x-t)_+^n}{n!}\right) = \frac{1}{n!} R_n((x-t)_+^n) = \frac{1}{n!} ((x-t)_+^n - P_n((x-t)_+^n)),$$

$$(x-t)_+^n = ((x-t)_+)^n = \begin{cases} (x-t)^n, & x \geq t \\ 0, & x < t \end{cases}.$$

- If K has constant sign on $[a, b]$, then (by the MVT)

$$(R_n(f))(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} R_n(e_{n+1})(x), \quad \xi \in [a, b].$$

- The function $F(x) = (x-t)_+^n$ has the **derivative**

$$F'(x) = \frac{\partial[(x-t)_+^n]}{\partial x} = n(x-t)_+^{n-1}$$

and the **integral** (this will only be needed later on, in Chapter 4)

$$\int_a^b F(x) dx = \frac{1}{n+1} (x-t)_+^{n+1} \Big|_{x=a}^{x=b} = \frac{1}{n+1} [(b-t)_+^{n+1} - (a-t)_+^{n+1}]$$

- Peano's theorem can also be used to derive remainder formulas for Lagrange or Hermite interpolation, but we already have other forms for those (recall that Peano's theorem refers to *any* linear functional).

III. Practice Problems

Find the polynomial of minimum degree that interpolates the given data. Determine the remainder for each approximation.

1. $f(0), f(1/2), f(1)$; **Answer:**

$$L_2 f(x) = (2x-1)(x-1)f(0) + 4x(1-x)f(1/2) + x(2x-1)f(1),$$

$$R_2 f(x) = \frac{x(x-1/2)(x-1)}{3!} f'''(\xi), \quad \xi \in (0, 1).$$

2. $f(-1), f(0), f'(0), f(1)$; **Answer:**

$$H_3 f(x) = \frac{1}{2} x^2 (1-x) f(-1) + (1-x^2) f(0) + x(1-x^2) f'(0) + \frac{1}{2} x^2 (x+1) f(1),$$

$$R_3 f(x) = \frac{x^2(x^2-1)}{4!} f^{(iv)}(\xi), \quad \xi \in (-1, 1).$$

3. $f(0), f'(0), f'(1)$. **Answer:**

$$B_2 f(x) = f(0) + \frac{1}{2} x(2-x) f'(0) + \frac{1}{2} x^2 f'(1),$$

$$R_2 f(x) = \frac{2x^3 - 3x^2}{2 \cdot 3!} f'''(\xi), \quad \xi \in (0, 1).$$

Lab Nr. 9, Numerical Calculus

Cubic Spline Interpolation; Least Squares Approximation

Matlab functions

- *spline* and *pchip*
- *polyfit* and *polyval*

Applications

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \frac{x+1}{3x^2+2x+1}.$$

Plot on the same set of axes the nodes, f and the Lagrange, Hermite (with double nodes) and deBoor spline interpolants at 7 equally spaced nodes in the interval $[-2, 4]$.

2. Consider the function $f(x) = x \sin(\pi x)$ and the nodes $\{-1, -1/2, 0, 1/2, 1, 3/2\}$.
- a) Obtain the deBoor, complete and piecewise Hermite cubic splines of these data;
 - b) Plot the nodes, f and the three spline functions, on the same set of axes, for $x \in [-1, 3/2]$.

3. The following data is given:

x	0.5	1.5	2	3	3.5	4.5	5	6	7	8
f	5	5.8	5.8	6.8	6.9	7.6	7.8	8.2	9.2	9.9

- a) Draw the scatterplot and find the least squares polynomial approximation that best fits the data;
 - b) Compute the error of this approximation;
 - c) Use this approximation to estimate the value at $x = 4$;
 - d) Plot on the same graph the data and the approximation polynomial.
4. The following table contains values of the pressure P of water vapors (measured in bars) as a function of temperature T (measured in $^{\circ}\text{C}$):

T	0	10	20	30	40	60	80	100
P	0.0061	0.0123	0.0234	0.0424	0.0738	0.1992	0.4736	1.0133

- a) Find the quadratic and cubic least squares approximation polynomials of these data. Which approximation is better?
- b) Approximate the pressure corresponding to a temperature of $T = 45^{\circ}$;
- c) Plot on the same set of axes the data and the two approximating polynomials.

Optional

5. Implement the computation of the *natural* cubic spline approximation. Then plot it together with the complete spline for the function and nodes in Problem 2.

Lab Nr. 10, Numerical Calculus

Numerical Integration I

Newton-Cotes Formulas; Adaptive Quadratures

Matlab functions

- *int*: computes definite and indefinite integrals symbolically
- *integral*: computes definite integrals numerically

1. Implement the composite rectangle, trapezoidal and Simpson's formulas.
2. Implement adaptive quadratures based on the three rules above.

Applications

1. Approximate $\ln 2$ with 3 correct decimals, applying the repeated rectangle, trapezoidal and Simpson's rules to the integral

$$I = \int_1^2 \frac{dx}{x},$$

with the appropriate number of subintervals. Compute the errors of the approximations.

2. Recall from Calculus that for a function $f : [a, b] \rightarrow \mathbb{R}_+$, the *area* under the curve $y = f(x)$ is given by the integral

$$A(f) = \int_a^b f(x) dx.$$

- a) Plot the graph of the function

$$f(x) = \frac{xe^{-x}}{x^2 + 1}, \quad x \in [0, 1].$$

- b) Approximate the area under f , using all three composite rules mentioned above, for $n = 2, 4, 8, 16, \dots, 256$. Compute the errors and the ratios of errors. Are the rates of convergence consistent with the theoretical ones?

3. The **error function** (also known as the **Gauss error function**), defined as

$$\operatorname{erf}(x) = \frac{2}{\pi} \int_0^x e^{-t^2} dt, \quad x \geq 0,$$

is an important function in mathematics, with applications in expressing solutions of the *heat equation*, in computing the cumulative distribution function (CDF) of the Normal distribution, and many more. Use adaptive trapezoidal quadratures to tabulate its values for $x = 0.1, 0.2, \dots, 1$. Compare the results with those given by Matlab functions *integral* and *erf*.

4. Recall from Calculus that the *length* of the curve represented by a differentiable function $y = f(x)$ on an interval $[a, b]$, is given by the integral

$$l(f) = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx.$$

- a) Plot the graph of the function $f : [0, 1] \rightarrow \mathbb{R}$, $f(x) = \sin(\pi x)$.
- b) Use adaptive Simpson quadratures to approximate the length of the curve.

Optional

5. The theoretical rates of convergence for the rectangle, trapezoidal and Simpson's rules assume a certain degree of smoothness of the integrand ($f \in C^2[a, b]$ for the first two and $f \in C^4[a, b]$ for the last one). These formulas *will not* perform as well for functions f that do not satisfy those conditions, in which case, other procedures might be necessary for numerically integrating them. Here is such an example.

- a) Apply Simpson's rule to the integral

$$I = \int_0^1 \sin(\sqrt[3]{x}) \, dx,$$

with $n = 2, 4, 8, \dots, 256$. Compute the errors and their ratios. What can be noticed?

- b) Transform I by using a suitable change of variables that will remove the singularity of the derivatives of the integrand and repeat part a).

Lab Nr. 11, Numerical Calculus

Numerical Integration II

Romberg's Method; Gaussian Quadratures

1. Implement Romberg's method for the trapezoidal rule. Keep also track of the number of function evaluations.
2. For the five weight functions and intervals given in the table, implement a Matlab routine that uses Gaussian quadratures to approximate the value of an integral, with a given number of nodes. Compute the nodes and the coefficients of the Gaussian quadrature formula using the Jacobi matrix (see Theorem 3.18, Lecture 10).

Applications

1. Approximate the integral

$$I = \int_0^{\pi/2} \frac{dx}{2 + \sin x}$$

(whose exact value is $\frac{\pi\sqrt{3}}{9}$), with 5 correct decimals using Romberg's method for the trapezium rule. Compute the error of the approximation and the number of function evaluations.

2. Approximate π using a suitable Gaussian quadrature with 2 nodes. Find the error of the approximation.

3. Approximate the integral

$$I = \int_0^{\pi/4} e^{\cos x} dx,$$

using a Gaussian formula with $n = 1, 2, \dots, 5$ nodes. Display the approximate values of I and the errors of each approximation.

4. Use appropriate Gaussian quadratures with $n = 2, 4, 6, 8$ and 10 nodes to approximate the integrals

a) $\int_0^{\infty} e^{-x} \sin x \, dx \left(= \frac{1}{2} \right);$

b) $\int_{\mathbb{R}} e^{-x^2} \cos x \, dx;$

c) $\int_{-1}^1 \frac{\sin(x^2)}{\sqrt{1-x^2}} \, dx.$

Optional

5. Consider the integral

$$I = \int_0^1 \frac{\sin x}{x} dx.$$

- a) Compute I with the Matlab function *integral*. So the integral exists.
- b) Use Romberg's method for the trapezoidal rule to approximate I . What is happening and why? Find a solution and fix the problem.

Name	Notation	Polynomial	Weight Fn.	Interval	α_k	β_k
Legendre	l_m	$[(x^2 - 1)^m]^{(m)}$	1	$[-1, 1]$	0	$\beta_0 = 2,$ $\beta_k = (4 - k^{-2})^{-1}, k \geq 1$
Chebyshev 1 st	T_m	$\cos(m \arccos x)$	$(1 - x^2)^{-\frac{1}{2}}$	$[-1, 1]$	0	$\beta_0 = \pi,$ $\beta_1 = \frac{1}{2},$ $\beta_k = \frac{1}{4}, k \geq 2$
Chebyshev 2 nd	Q_m	$\frac{\sin[(m+1) \arccos x]}{\sqrt{1-x^2}}$	$(1 - x^2)^{\frac{1}{2}}$	$[-1, 1]$	0	$\beta_0 = \frac{\pi}{2},$ $\beta_k = \frac{1}{4}, k \geq 1$
Laguerre	L_m^a	$x^{-a} e^x (x^{m+a} e^{-x})^{(m)}$	$x^a e^{-x}, a > -1$	$[0, \infty)$	$2k + a + 1$	$\beta_0 = \Gamma(1 + a),$ $\beta_k = k(k + a), k \geq 1$
Hermite	H_m	$(-1)^m e^{x^2} (e^{-x^2})^{(m)}$	e^{-x^2}	$(-\infty, \infty)$	0	$\beta_0 = \sqrt{\pi},$ $\beta_k = \frac{k}{2}, k \geq 1$

Lab Nr. 12, Numerical Calculus

Numerical Methods for Nonlinear Equations

Matlab functions

- *solve*: solves nonlinear equations and systems symbolically
- *fsolve*: solves nonlinear equations and systems numerically
- *roots*: finds roots of polynomials
- *factor*: finds factors of numbers or symbolic expressions

Implement the bisection, secant and Newton's methods for approximating numerical solutions of a nonlinear equation $f(x) = 0$, with a given precision. Keep also track of the number of iterations needed for a desired precision.

Applications

1. Let $f(x) = x - e^{-x}$.

- a) Locate the real roots of the equation $f(x) = 0$ analytically and graphically.
- b) Approximate the real roots of the equation using the three methods above, with appropriate starting values. Compare the speed of convergence of the three methods.

2. Consider the polynomial equation

$$p(x) \equiv x^4 + 2x^3 - 4x^2 - 14x - 5 = 0.$$

- a) Use Matlab functions *roots*, *solve* and *fsolve* to find all (real and complex) roots of p .
- b) Factor p and plot its graph for $x \in [-1, 3]$.
- c) Approximate all (real and complex) roots of p , using the three methods above.
- d) Use the secant method with values $x_0 = -3 - 2i$, $x_1 = -1 - i$, then Newton's method with $x_0 = 3 + i$. What do you notice?

3. Let

$$f(x) = x + e^{-Bx^2} \cos x, \text{ for some parameter } B > 0.$$

- a) Show that the equation $f(x) = 0$ has a real root in the interval $(-1, 0)$, for any $B > 0$ (it can be shown that that is the *only* root). For $B = 1, 5, 50$ and 100 , plot the function f for $x \in [-1.2, 0.5]$.
- b) For $B = 1$ and 5 , approximate the root using Newton's method with starting value $x_0 = 0$.
- b) Can you do the same for $B = 50$ and 100 ? Explain what is happening and find a way to approximate the root using Newton's method.

Optional

4. An interesting rootfinding problem occurs in the computation of *annuities* (which are savings accounts or trust funds). Suppose you are paying into an account an amount of P_{in} per period of time, for N_{in} periods of time. The amount you deposit is compounded at an interest rate of r per period of time. Then at the beginning of period $N_{in} + 1$, you will withdraw an amount of P_{out} per time period, for N_{out} periods. It is assumed that the interest rate r holds over all $N_{in} + N_{out}$ periods. In order that the amount you withdraw balance that which has been deposited, including interest, what is the needed interest rate? The mathematical model of this problem leads to the equation

$$P_{in} \left[(1 + r)^{N_{in}} - 1 \right] = P_{out} \left[1 - (1 + r)^{-N_{out}} \right].$$

As a particular case, suppose you are paying in $P_{in} = 1000$ currency units (CU = RON, EUR, USD, etc.) each month for 40 years. Then you wish to withdraw $P_{out} = 5000$ CU's per month for 20 years. If the interest rate is R per year, compounded monthly, then $r = R/12$. Also, $N_{in} = 40 \cdot 12 = 480$ and $N_{out} = 20 \cdot 12 = 240$. Thus, we want to solve

$$f(r) \equiv 1000 \left[\left(1 + \frac{R}{12} \right)^{480} - 1 \right] - 5000 \left[1 - \left(1 + \frac{R}{12} \right)^{-240} \right] = 0.$$

What is the needed yearly interest rate R ?

- a) Plot the graph of f on an appropriate interval and locate the real roots of f .
- b) Approximate the positive root, using the bisection and secant methods with various starting values. Compare the results. Would Newton's method be appropriate in this case?

Note: This example shows the power of *compound* interest (compound interest is when you earn interest on the money you've saved *and* on the interest you earn along the way).

7. Lagrange interpolation in Newton's divided differences form

 $f: [0, 1] \rightarrow \mathbb{R}, f(x) = \cos(\pi x)$

x	0	$1/3$	$1/2$	1
$f(x)$	1	$1/2$	0	-1

$$m = 3$$

$$\Downarrow$$

$$\boxed{n = 3}$$

(a) Lagrange polynomial

$$L_3 f(x) = N_3 f(x) = f(x_0) + f[x_0, x_1](x-x_0) \\ + f[x_0, x_1, x_2](x-x_0)(x-x_1) \\ + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2)$$

x	$f(x)$	$D^1 f$	$D^2 f$	$D^3 f$
0	1	$-3/2$	-3	$9/2$
$1/3$	$1/2$	-3	$3/2$	
$1/2$	0	-2		
1	-1			

$$D^i f(x_0) = f[x_0, \dots, x_i]$$

Hence, $L_3 f(x) = 1 - \frac{3}{2}(x-0) - 3(x-0)(x-\frac{1}{3}) + \frac{9}{2}(x-0)(x-\frac{1}{3})(x-\frac{1}{2})$

$$= 1 - \frac{3}{2}x - \frac{3x(3x-1)}{3} + \frac{9x(3x-1)(2x-1)}{12}$$

$$\Rightarrow L_3 f(x) = 1 - \frac{3}{2}x - x(3x-1) + \frac{3x(3x-1)(2x-1)}{4}$$

$$= \frac{9}{2}x^3 - \frac{27}{4}x^2 + \frac{1}{4}x + 1$$

b) Approx. for $\cos(\frac{\pi}{5})$

$$\cos(\frac{\pi}{5}) = f(\frac{1}{5}) \approx L_3 f(\frac{1}{5}) ,$$

$$\text{where } L_3 f(\frac{1}{5}) = \frac{9}{2}(\frac{1}{5})^3 - \frac{27}{4}(\frac{1}{5})^2 + \frac{1}{4} \cdot \frac{1}{5} + 1 = \dots = 0,816$$

c) The error $R_3 f$

$$R_3 f(x) = \frac{u(x)}{(3+1)!} f^{(4)}(\xi) , \text{ where } \xi \in (0, 1)$$

$$f(x) = \cos(\pi x) \Rightarrow f'(x) = -\pi \sin(\pi x)$$

$$f''(x) = -\pi^2 \cos(\pi x)$$

$$f'''(x) = \pi^3 \sin(\pi x)$$

$$f^{(4)}(x) = \pi^4 \cos(\pi x)$$

$$\text{and } u(x) = (x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$$= (x-0)(x-\frac{1}{3})(x-\frac{1}{2})(x-1)$$

$$\text{Then } R_3 f(x) = \frac{x(3x-1)(2x-1)(x-1)}{6 \cdot 4!} \pi^4 \cos(\pi \cdot \xi) ,$$

$$\xi \in (0, 1) .$$

d) A bound for the error

$$\begin{aligned}
 0 \leq |R_3 f(x)| &= \left| \frac{x(3x-1)(2x-1)(x-1)}{6 \cdot 4!} \pi^4 \cos(\pi \cdot x) \right| \\
 &\leq \frac{|x(3x-1)(2x-1)(x-1)|}{6 \cdot 24} \cdot \pi^4 \cdot \underbrace{|\cos(\pi \cdot x)|}_{\leq 1} \\
 &\leq \frac{\pi^4}{144} \cdot \underbrace{\left| \frac{v(x)}{x(3x-1)(2x-1)(x-1)} \right|}_{\leq \max_{x \in [0,1]} v(x)}, \quad x \in [0,1]
 \end{aligned}$$

• In this case, v is of degree 4 and is difficult to determine by hand the maximum value (see Lecture 5 - pag. 7 for an example)

e) A bound for the error of approx. $f(\frac{1}{5})$

$$0 \leq |R_3 f(\frac{1}{5})| \leq \frac{\pi^4}{6 \cdot 24} \cdot \frac{1}{5} \cdot \frac{2}{5} \cdot \frac{3}{5} \cdot \frac{4}{5} = \frac{\pi^4}{5^4 \cdot 6} = \frac{1}{6} \left(\frac{\pi}{5}\right)^4$$

$$\Rightarrow 0 \leq |R_3 f(\frac{1}{5})| \leq \frac{1}{6} \cdot \left(\frac{\pi}{5}\right)^4 \approx 0.0259$$

$$\text{Indeed, } |L_3 f(\frac{1}{5}) - f(\frac{1}{5})| = 0.00698 < 0.0259$$

□

8. Polynomial Interpolation - summary

LAGRANGE

- classic
- barycentric
- Aitken / Neville

HERMITE
DOUBLE
NODES

HERMITE

BIRKHOFF

fundamental (basis)
polynomials

Newton's form can be
applied to both types
(+ divided differences)

Schematic example

Lagrange

x	x_0	x_1
f	$f(x_0)$	$f(x_1)$
f'	X	X

Hermite (double nodes)

x	x_0	x_1
f	$f(x_0)$	$f(x_1)$
f'	$f'(x_0)$	$f'(x_1)$

Hermite classic

x	x_0	x_1	x_2
f	$f(x_0)$	$f(x_1)$	$f(x_2)$
f'	X	$f'(x_1)$	$f'(x_2)$
f''	X	$f''(x_1)$	X

Birkhoff (incomplete
Hermite)

x	x_0	x_1	x_2
f	$f(x_0)$	X	$f(x_2)$
f'	X	$f'(x_1)$	X
f''	X	X	$f''(x_2)$

Exercises

 $f: (-1, \infty) \rightarrow \mathbb{R}, f(x) = \frac{1}{1+x}$

$$x_0 = 0, x_1 = 1, x_2 = 2$$

x	0	1	2
$f(x)$	1	$1/2$	$1/3$

A. Lagrange

A.1. Classic : $m = \text{degree of the polynomial}$
 $= \text{number of nodes} - 1$

(e.g. $\underbrace{x_0, x_1, \dots, x_m}_{(m+1) \text{ nodes}} \Rightarrow m = m$)

Here, $\boxed{m=2} \Rightarrow L_2 f(x) = \sum_{i=0}^2 l_i(x) f(x_i)$

$$\begin{aligned} \Rightarrow L_2 f(x) &= l_0(x) f(x_0) + l_1(x) f(x_1) + l_2(x) f(x_2) \\ &= 1 \cdot l_0(x) + \frac{1}{2} \cdot l_1(x) + \frac{1}{3} \cdot l_2(x) \end{aligned}$$

$$\bullet l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-1)(x-2)}{(0-1)(0-2)} = \frac{(x-1)(x-2)}{2}$$

$$\bullet l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-0)(x-2)}{(1-0)(1-2)} = -x(x-2)$$

$$\bullet l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-0)(x-1)}{(2-0)(2-1)} = \frac{x(x-1)}{2}$$

Then

$$L_2 f(x) = \frac{1}{(x-1)(x-2)} - \frac{1}{2} \frac{x(x-2)}{2} + \frac{x(x-1)}{6}$$

$$= \frac{3x^2 - 9x + 6 - 3x^2 + 6x + x^2 - x}{6} = \frac{1}{6}x^2 - \frac{2}{3}x + 1$$

$$\Rightarrow f(x) \approx L_2 f(x) = \frac{1}{6}x^2 - \frac{2}{3}x + 1$$

A.2. Newton form : $n = m$

$$N_2 f(x) = f(x_0) + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$$

$\underbrace{\hspace{10em}}_{x^2 = \text{degree 2}}$

$\underbrace{\hspace{10em}}_{\text{divided differences}}$

$$= f(0) + f[x_0, x_1]x + f[x_0, x_1, x_2]x(x-1)$$

x	$f(x)$	$\mathcal{D}^1 f$	$\mathcal{D}^2 f$
0	1	$f[x_0, x_1] = \frac{1/2 - 1}{1 - 0} = -1/2$	$f[x_0, x_1, x_2] = \frac{-1/6 + 1/2}{2 - 0} = 1/6$
1	1/2	$f[x_1, x_2] = \frac{1/3 - 1/2}{2 - 1} = -1/6$	
2	1/3		

Hence, $N_2 f(x) = 1 - \frac{1}{2}x + \frac{1}{6}x(x-1) = \frac{1}{6}x^2 - \frac{2}{3}x + 1$

3. Hermite (double nodes)

$$f(x) = \frac{1}{1+x} = (1+x)^{-1}$$

$$\Rightarrow f'(x) = [(1+x)^{-1}]' = -1 \cdot (1+x)^{-2} \cdot \underbrace{(1+x)'}_1 = \frac{-1}{(1+x)^2}$$

x	0	1	2
f(x)	1	1/2	1/3
f'(x)	-1	-1/4	-1/9

Pattern: time - x
distance - f
speed - f'

- In this case, $m = \text{degree of the polynomial}$

$$m = 2 \cdot n + 1$$

Here, $x_0 = 0$
 $x_1 = 1 \Rightarrow n = 2 \Rightarrow m = 2 \cdot 2 + 1 = 5$
 $x_2 = 2$

$$m = 5$$

- We apply the Newton's form for Hermite polynomial with double nodes

$$N_5 f(z) = f(z_0) + \sum_{i=1}^5 (z-z_0) \cdots (z-z_{i-1}) D^i f(z_0)$$

$$\begin{aligned} &= f(z_0) + f[z_0, z_1](z-z_0) + f[z_0, z_1, z_2](z-z_0)(z-z_1) \\ &+ f[z_0, \dots, z_3](z-z_0)(z-z_1)(z-z_2) + \\ &+ f[z_0, \dots, z_4](z-z_0) \cdots (z-z_3) \\ &+ f[z_0, \dots, z_5](z-z_0) \cdots (z-z_4) \end{aligned}$$

z	$f(z)$	$D^1 f$	$D^2 f$	$D^3 f$	$D^4 f$	$D^5 f$
$x_0 \begin{cases} 0 = z_0 \\ 0 = z_1 \end{cases}$	1 1	(-1) $-1/2$	$1/2$ $1/4$	$-1/4$ $-1/12$	$1/12$ $1/36$	$-1/36$
$x_1 \begin{cases} 1 = z_2 \\ 1 = z_3 \end{cases}$	$1/2$ $1/2$	$(-1/4)$ $-1/6$	$1/12$ $1/18$	$-1/36$		
$x_2 \begin{cases} 2 = z_4 \\ 2 = z_5 \end{cases}$	$1/3$ $1/3$	$(1/9)$				

↑ derivatives

- In this table, $f(z_0) = 1$
 $f[z_0, z_1] = -1 = D^1 f(z_0)$
 $f[z_0, z_1, z_2] = 1/2 = D^2 f(z_0)$
 $\vdots$

$$\begin{aligned}
 \Rightarrow H_5 f(z) = N_5 f(z) &= 1 - 1(z-0) + \frac{1}{2}z(z-0) - \\
 &\quad - \frac{1}{4}z^2(z-1) + \frac{1}{12}z^2(z-1)^2 - \frac{1}{36}z^2(z-1)^2(z-2) \\
 &= \frac{-z^2(z-1)^2(z-2)}{36} + \frac{z^2(z-1)^2}{12} - \frac{z^2(z-1)}{4} + \frac{z^2}{2} - z + 1 \\
 &= -\frac{1}{36}z^5 + \frac{7}{36}z^4 - \frac{5}{9}z^3 + \frac{8}{9}z^2 - z + 1
 \end{aligned}$$



t (time)	0	3	5	8	13
$f(t)$ = distance	0	225	383	623	993
$f'(t)$ = speed	0	77	80	74	72

5 modes
 $\Downarrow$
 $m = 4$
 $\Downarrow$
 $m = 9$

3 Example for Birkhoff (in particular, for Hermite classic)

$$\begin{array}{lcl} x_0 = 0 & , & f(0) = 2 \quad , \quad f'(0) = 1 \\ x_1 = 1 & , & \text{---} \quad , \quad f'(1) = 3 \end{array}$$

($f(1)$ is unknown $\Rightarrow$ Birkhoff problem)

3.1. Direct method

$$\begin{aligned} m &= \text{degree of the polynomial} \\ &= |I_0| + \dots + |I_m| - 1 \end{aligned}$$

(I_k is a set associated to x_k)

$\hookrightarrow$ contains the orders of the derivatives for x_k ,

where $f(x_k) = f^{(0)}(x_k)$ is the derivative of order 0)

• In our case ,

$$\begin{array}{lcl} I_0 & = & \{0, 1\} \\ I_1 & = & \{1\} \end{array}$$

Arrows indicate: $I_0 \rightarrow f(x_0)$ and $f'(x_0)$; $I_1 \rightarrow f'(x_1)$

• $|I_0| = \text{card}(I_0) = \text{number of elements}$

$$\Rightarrow m = |I_0| + |I_1| - 1 = 2 + 1 - 1 = 2$$

$$\boxed{m = 2}$$

so we have a Birkhoff polynomial of degree 2

$$\mathcal{B}_2 f(x) = ax^2 + bx + c \quad (\text{the standard form})$$

$$(\mathcal{B}_2 f)'(x) = 2ax + b$$

We have that $\mathcal{B}_2 f(x_0) = f(x_0)$

~~$\mathcal{B}_2 f(x_1) = f(x_1)$~~ is unknown

$$(\mathcal{B}_2 f)'(x_0) = f'(x_0)$$

$$(\mathcal{B}_2 f)'(x_1) = f'(x_1)$$

$$\Rightarrow \mathcal{B}_2 f(0) = f(0) \Leftrightarrow c = f(0) = 2$$

$$(\mathcal{B}_2 f)'(0) = f'(0) \Leftrightarrow b = f'(0) = 1$$

$$(\mathcal{B}_2 f)'(1) = f'(1) \Leftrightarrow 2a + b = f'(1) = 3$$

$$\Leftrightarrow 2a + f'(0) = f'(1)$$

$$\Leftrightarrow a = \frac{1}{2}[f'(1) - f'(0)] = 1$$

Hence,

$$\mathcal{B}_2 f(x) = \frac{1}{2}[f'(1) - f'(0)]x^2 + f'(0) \cdot x + f(0)$$

$$= f(0) + \frac{1}{2}x(2-x)f'(0) + \frac{1}{2}x^2 f'(1)$$

$$= 1 \cdot \underbrace{f(0)}_{\text{fundamental (basis) polynomials}} + \frac{1}{2}x(2-x) \underbrace{f'(0)}_{\text{fundamental (basis) polynomials}} + \frac{1}{2}x^2 \underbrace{f'(1)}_{\text{data}}$$

fundamental (basis)
polynomials

data

In our example, $B_2 f(x) = 1 \cdot x^2 + 1 \cdot x + 2 = x^2 + x + 2$

3.2. Basis (fundamental) polynomials

- this method gives us a way to determine the basis polynomials (one by one)

$$B_m f(x) = \sum_{k=0}^m \sum_{j \in I_k} b_{kj}(x) f^{(j)}(x_k)$$

Here,

$$B_2 f(x) = \sum_{k=0}^1 \sum_{j \in I_k} b_{kj}(x) f^{(j)}(x_k)$$

$$= \underbrace{b_{00}(x)f(x_0) + b_{01}(x)f'(x_0)}_{\substack{k=0 \\ j \in I_0 = \{0, 1\}}} + \underbrace{b_{11}(x)f'(x_1)}_{\substack{k=1 \\ j \in I_1 = \{1\}}}$$

$$= \underline{b_{00}(x)} \underline{f(0)} + \underline{b_{01}(x)} \underline{f'(0)} + \underline{b_{11}(x)} \underline{f'(1)}$$

Since $B_2 f$ is of degree 2, we start with basis polynomials of the same degree.

! All the data we know for f is transferred to the basis polynomials.

$$(i) \quad b_{00}(x) = ax^2 + bx + c$$

$$b'_{00}(x) = 2ax + b$$

We know $\begin{cases} b_{00}(x_0) \\ b'_{00}(x_0) \\ b'_{00}(x_1) \end{cases}$ and based on the rule

$$b_{kj}^{(j)}(x_k) = 1, \text{ otherwise } 0$$

$$\text{we obtain } \begin{cases} b_{00}(x_0) = \boxed{1} \\ b'_{00}(x_0) = 0 \\ b'_{00}(x_1) = 0 \end{cases} \Leftrightarrow \begin{cases} c = 1 \\ b = 0 \\ 2a + b = 0 \end{cases}$$

$$\Rightarrow \boxed{b_{00}(x) = 1}$$

$$(ii) \quad b_{01}(x) = ax^2 + bx + c$$

$$\begin{cases} b_{01}(x_0) = 0 \\ b'_{01}(x_0) = \boxed{1} \\ b'_{01}(x_1) = 0 \end{cases} \Leftrightarrow \begin{cases} c = 0 \\ b = 1 \\ a = -1/2 \end{cases} \Rightarrow \boxed{b_{01}(x) = \frac{1}{2}x(2-x)}$$

$$(iii) \quad b_{11}(x) = ax^2 + bx + c$$

$$\begin{cases} b_{11}(x_0) = 0 \\ b'_{11}(x_0) = 0 \\ b'_{11}(x_1) = \boxed{1} \end{cases} \Leftrightarrow \begin{cases} c = 0 \\ b = 0 \\ a = 1/2 \end{cases} \Rightarrow \boxed{b_{11}(x) = \frac{1}{2}x^2}$$

$$\text{Hence, } B_2 f(x) = 1 \cdot f(0) + \frac{1}{2}x(2-x)f'(0) + \frac{1}{2}x^2 f'(1)$$

□